

SUMMARY

Full-stack software developer with experience building scalable backend systems, real-time distributed applications, and AI-powered solutions using Node.js, React, React Native, GraphQL, and modern databases. Passionate about backend engineering, distributed systems, and Generative AI, with industry experience spanning enterprise software development and AI consulting.

EDUCATION

- Indian Institute of Technology, Dharwad
B.Tech, Computer Science and Engineering
Aug 2023 – Jul 2027
CGPA: 8.81 (Present)
- Senior Secondary Education (Class XII)
Board of Intermediate Education, Andhra Pradesh
Apr 2022 – Apr 2023
Percentage: 96.3%
- Secondary Education (Class X)
Indian Certificate of Secondary Education (ICSE)
Apr 2020 – Apr 2021
Percentage: 96.2%

WORK EXPERIENCE

- Advanced Application Engineering Analyst Intern – Accenture
Chennai, Tamil Nadu (On-site)
May 2026 – Jul 2026
 - Conducted market research on enterprise **Agentic AI** adoption in the **Japanese market**, authoring a **technical white paper** analyzing buyer demand, AI governance, cloud ecosystems, and deployment opportunities across healthcare, manufacturing, BFSI, retail, and government sectors.
 - Built a prototype called **ElderCareAI**, a **multi-agent AI system** for Japanese elder care facilities, orchestrating specialized agents for ML-based health anomaly detection, fall-risk monitoring, caregiver scheduling, and LLM-generated caregiver summaries using **Ollama + LLaMA2**, with an offline-first architecture for edge deployment.
 - Identified **white-space opportunities** for Agentic AI adoption across multiple industries and proposed domain-specific enterprise solutions and investment opportunities, enabling Accenture to evaluate potential AI service offerings for the Japanese market.

- SDE Intern – TP Vision
Bengaluru, Karnataka (On-site)
May 2025 – Jul 2025
 - Developed scalable **RESTful APIs** using **Fastify** and **TypeBox** with **MongoDB** and **PostgreSQL**, implementing a **hybrid backend architecture** integrated with **GraphQL** for efficient data retrieval; developed responsive, component-driven interfaces for the **Admin Portal** using **React.js**.
 - Designed and implemented the **Unclaim** workflow within the Admin Portal, enabling secure revocation and reassignment of display licenses and application subscriptions for deactivated devices across customer accounts.
 - Developed a centralized display management interface in the Admin Portal to streamline device monitoring, management, and operational workflows.
 - Integrated **AWS Simple Email Service (SES)** for secure transactional email delivery, enabling authentication and authorization of the Unclaim workflow while reducing manual intervention.

PROJECTS

- Wellness Centre Notification System
Real-time IoT nurse-call system | Node.js · Socket.IO · React Native · Raspberry Pi · PM2
Jan 2026
 - Designed and developed a **real-time bidirectional communication** backend using **Node.js**, **Express.js**, and **Socket.IO**, serving three concurrent client types - React Native mobile app, nurse station display, and reception TV - with sub-second event propagation across all connected clients.
 - Architected a **REST + WebSocket hybrid API**: a POST /api/login endpoint handles **role-based authentication** (doctor/staff), while Socket.IO events (doctorCall, completeCall, clearAllCalls) manage the complete real-time call lifecycle across all displays.
 - Implemented **network-layer access control** using **CIDR-based IP validation** configured via environment variables, blocking all out-of-network requests before authentication - eliminating unauthorized external access without requiring a VPN.
 - Deployed the system on a **Raspberry Pi** display server; authored shell startup scripts and configured **PM2** process manager to auto-launch nurse and reception web displays on boot, achieving **zero-touch operation** post-setup.

- DataFort - Blockchain-Based Secure File Storage
Distributed file storage system | Node.js · Express.js · MongoDB · IPFS · Exonum Blockchain
Oct 2024
 - Developed the complete **Node.js/Express.js** backend, implementing all server-side routes for user registration, login, file upload, and file retrieval, serving as the core application layer between the client, MongoDB, and the distributed IPFS/Exonum storage stack.
 - Engineered the **file ingestion pipeline**: on upload, the backend streams the file to a local **IPFS** node, receives the returned content hash (**CID**), and persists the CID alongside file metadata to **MongoDB** - decoupling content addressing from the application database.
 - Implemented **session-based authentication** with hashed credential validation against MongoDB, securing all upload and retrieval endpoints behind auth middleware and enforcing per-user **role-based access control** on stored files.
 - Integrated the **Exonum** private permissioned blockchain to record each file transaction as an immutable ledger entry, providing a **tamper-evident audit trail** fully independent of the application database.

- Developed the complete **Node.js/Express.js** backend, implementing all **RESTful API** routes for book issuance, returns, reservations, and availability queries, with session middleware enforcing **role-based access control** between admin and student accounts.
- Designed the **MySQL** relational schema covering users, book inventory, reservation records, and transaction history; wrote optimized queries for real-time availability checks, overdue detection, and per-user borrow-limit enforcement.
- Integrated EJS server-side templating to dynamically render role-specific dashboards, injecting live session state and query results into distinct admin and student views without client-side framework overhead.

TECHNICAL SKILLS AND COURSES

- **Languages**: C, C++, Java, JavaScript, Python
- **Frameworks**: Node.js, Express.js, React.js, GraphQL, Fastify
- **Databases**: MongoDB, PostgreSQL, MySQL, Supabase, SQLite
- **AI/ML Libraries/Frameworks**: scikit-learn, Ollama, LLaMA2, pandas, Flask
- **Tools**: Git, GitHub, Postman, Docker, PM2, AWS SES, MongoDB Compass
- **Courses**: Data Structures & Algorithms, Computer Architecture, Operating Systems, DBMS, Computer Networks, Machine Learning, Deep Learning, Linear Algebra, Data Analysis

POSITIONS OF RESPONSIBILITY & VOLUNTEERING

- | | |
|---|---------------------|
| • Student Mentor , Student Mentorship Program, IIT Dharwad | Jul 2024 – Aug 2025 |
| • Member , Robotics Club, IIT Dharwad | Aug 2023 – Aug 2024 |
| • Student Volunteer , National Service Scheme, IIT Dharwad | Aug 2023 – Dec 2023 |
| • Volunteer , SPIC MACAY, IIT Dharwad | Oct 2024 |

ACHIEVEMENTS & CERTIFICATIONS

- | | |
|---|----------|
| • Secured First Place for College Library Management System at Coding Club's SOI event, IIT Dharwad. | Sep 2024 |
| • Finlatics Machine Learning Program , Finlatics | Mar 2025 |